

Children with MOG-IgG positive bilateral optic neuritis misdiagnosed as fulminant idiopathic intracranial hypertension

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ABSTRACT

Background: Fulminant idiopathic intracranial hypertension (IIH) is characterized by headache, rapid decrease of vision and elevated CSF-opening pressure.

Objective: To delineate a subgroup of MOGAD mimicking fulminant IIH.

Methods: In this case series children with MOGAD with vision loss, optic disc swelling and elevated CSF opening pressure, initially diagnosed with fulminant IIH, were included.

Results: 8 MOGAD patients (median age 12.5y, f:m = 7:1) with an initial diagnosis of fulminant IIH were included. Rapid bilateral visual loss and headache was present in all patients in addition to bilateral optic disc swelling and decreased visual acuity. In 2 patients cMRI showed prominent optic discs or an empty sella sign. In 7 patients the CSF opening pressure was markedly elevated. Patients were treated with Acetazolamide (n = 6), CSF external drainage (n = 2) or implantation of VP shunt (n = 1). Intravenous Methylprednisolone was administered in all patients because of worsening of symptoms, positive MOG-ab titers and/or new MRI white matter lesions. Visual recovery was good with residuals in 1/8 children. OCT revealed thickening of pRNFL in the acute stage in all patients with a rapid declining of pRNFL and retinal atrophy thereafter.

Conclusion: Children with bilateral MOGAD-ON presenting with loss of vision and elevated CSF opening pressure are at risk of being misdiagnosed as fulminant IIH. The role of elevated CSF opening pressure in MOGAD warrants further investigations. We recommend to include measurement of CSF opening pressure in the work up of neuroinflammatory diseases, in particular in patients with suspected MOGAD, to consider adapted pressure management.

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1. Introduction

Myelin-oligodendrocyte-glycoprotein-associated diseases (MOGAD) have become an important differential diagnosis in children with an acute demyelinating syndrome (ADS). Clinical presentation is age-dependent, with predominantly ADEM-like clinical and radiological presentation in young children. ON is one of the most frequent presentations of MOGAD and can occur unilateral and isolated or in combination with other subtypes. ON in MOGAD does occur also more often bilateral compared to MS or NMOSD and further ON episodes are a frequent form of recurrent MOGAD. (Ramanathan et al., 2018; Hacohen et al., 2018; Banwell et al., 2023; Wendel et al., 2020)

MOG-IgG positive ON is characterized by mild to severe unilateral and less often bilateral loss of vision frequently associated with optic disc swelling/severe papilledema of the affected eye. Patients with MOGAD show a good response to high dose steroids, often with complete resolution of symptoms. (Wendel et al., 2020) More recently it was shown that in particular children presenting with MOG-positive ADEM can have elevated cerebrospinal fluid (CSF) opening pressures requiring treatment usually reserved for children with pseudotumor cerebri syndrome (PTCS). (Nguyen et al., 2023)

PTCS is defined by an abnormally high intracranial pressure (CSF opening pressure > 28 cm H₂O) in the setting of signs of intracranial hypertension, including papilledema, a normal neurological examination (except for 6th nerve palsy) and no parenchymal lesions on neuroimaging. (Friedman et al., 2013) Common symptoms of PTCS include headaches and/or blurred vision, but the clinical presentation is highly variable, especially in children. (Matthews et al., 2017) A subgroup of PTCS presents with fulminant idiopathic intracranial hypertension (IIH). Fulminant IIH is defined by its acute onset with <4 weeks between symptom onset and severe vision loss. (Bouffard, 2020) Unlike in typical pediatric PTCS, severe visual loss occurs early and is often rapidly progressive. Affected patients are at high risk for permanent blindness if early including surgical treatment is delayed. (Thambisetty et al., 2007)

In this case series we report eight MOG antibody (ab) positive children presenting with acute loss of vision, optic disc swelling and an elevated CSF opening pressure, who were all initially diagnosed with fulminant IIH.

2. Methods

Children with a first ADS are referred to the BIOMARKER study (Wendel et al., 2022) by physicians from different European countries since 2009 until today. From this cohort of 1300 subjects we selected patients with MOGAD and bilateral ON who were initially diagnosed with fulminant IIH but subsequently referred to our study because they were finally found to have a bilateral ON in conjunction with serum MOG antibodies (abs). Eight patients from seven different medical centers fulfilled the following inclusion criteria including age < 18 years, initial diagnosis of fulminant IIH fulfilling the respective criteria (e.g. headache, signs of increased intracranial pressure such as bilateral papilledema (Friedman et al., 2013), initial treatment for IIH, positive test results for MOG-IgG abs and subsequent diagnosis of MOGAD. Diagnosis of MOGAD was based on the criteria recently published (Banwell et al., 2023): (A) a core clinical demyelinating event (in our cases: optic neuritis), (B) positive MOG-IgG test, (C) if MOG-IgG test positive with low titer or no reported titer: ≥ 1 supporting clinical or MRI feature (in our patients: bilateral simultaneous involvement and/or (subsequent) typical MRI features). Patients with a presentation suggestive of fulminant IIH, but with a prompt diagnosis of MOGAD were not included in this case series.

2.1. Optical coherence tomography (OCT)

Spectral-domain optical coherence tomography (SD-OCT, SPEC-TRALIS, Heidelberg Engineering, Heidelberg, Germany) with automatic

real-time (ART) averaging was used to measure the peripapillary retinal nerve fiber layer (pRNFL) thickness. Based on the standard protocol a ring scan of the optic nerve head with an activated eye tracker (12°, 3,5 mm ring, $50 \leq \text{ART} \leq 100$) has been conducted in all patients. Experienced evaluators carefully checked all scans for sufficient quality. The SD-OCT data were analyzed and reported according to the recommendations of APOSTEL2.0 and OSCAR-IB (Aytulun et al., 2021; Schippling et al., 2015). pRNFL atrophy was defined as a decrease below 1SD (pRNFL $G < 94.0 \mu\text{m}$), in comparison to a group of pediatric healthy controls ($n = 144$, f:m 32:40, age 12.49 ± 2.18). These healthy controls have been investigated in St. Josef-Hospital Bochum/Germany and were reported separately. (Runge et al., 2022)

2.2. Standard protocol approvals, registrations and patient consents

This BIOMARKER study was approved by the Ethics committee of Witten/Herdecke University, Germany as well as the Ethics committee of Medical University Innsbruck, Austria. Patients included via the BIOMARKER study gave written consent.

2.3. MOG antibody testing

Of four children included, serum samples were analyzed for the presence of MOG-antibodies by live cell-based immunofluorescence assays. MOG-IgG were tested using full-length MOG (alpha-1 isoform) and IgG (heavy and light chain, Dianova®) specific secondary abs. Screening was performed at dilutions of 1:20 and 1:40 by at least two independent clinically blinded investigators and positive serum samples were further diluted in two-fold increments to determine the end-point titers. Titer levels of $\geq 1:160$ were classified as MOG-IgG positive and confirmed using a second assay with an IgG(Fc) specific secondary antibody (Dianova®) as previously described. (Reindl et al., 2020) Seronegativity was defined as a MOG-IgG titer of <1:160. MOG titer were graded in clear positive titer (live CBA $\geq 1:640$; Euroimmune $\geq 1:100$) and low titer (live CBA 1:160–320; Euroimmune 1:10).

Serum from four children included were tested with fixed CBA-IF IgG (Fc): HEK293 are transiently transfected with a recombinant plasmid expression vector encoding human MOG X11 isoform and are fixed with formaldehyde. Screening was performed at 1:10 dilution, and bound MOG-IgG was detected using a fluorescein-labeled anti-human IgG(Fc) secondary antibody according to the manufacturer's instructions (Euroimmun).

3. Results

Eight patients with a median age of 12.5 years (range: 4–17y) were included in this retrospective case series, with seven female and one male patient. BMI ranged from 16.1kg/m² to 50kg/m² (median: 22.1kg/m²). Clinical data are summarized in Table 1. All patients had headache and visual impairment at initial presentation, in combination with optic disc swelling on ophthalmologic examination. In addition, patients had a uni- or bilateral abducens nerve palsy ($n = 2$), nausea/vomiting ($n = 4$), fatigue ($n = 2$) and vertigo ($n = 2$). In three patients, further signs of ON were reported (eye movement pain, $n = 2$; decreased color saturation, $n = 3$). CSF opening pressure was measured in all patients but invalid in patient 2 (measured in sitting position, 85 cm H₂O). In all other patients CSF opening pressure was elevated with a median of 43 cm H₂O (range: 34 to 60 cm H₂O). MOG-IgG testing subsequently showed positive results in all eight patients.

Eye examination revealed bilateral optic disc swelling in all patients, with peripapillary hemorrhages in three patients. Relative afferent pupillary defect (RAPD) was reported in three patients. Visual acuity was reduced in both eyes in all patients (VA range: 20/30 – 1/100), which solely allowed recognition of hand movement in most of them (Table 1). Visual evoked potentials (VEP) were performed in 6/8 patients, with pathological results in 5/6 children showing prolonged P100 latencies.

Table 1

Clinical and demographic data of initial presentation of nine children presenting with symptoms suggestive of fulminant idiopathic intracranial hypertension due MOG-associated disease.

	child – c, adolescent – a	CSF opening pressure (cm H ₂ O)	CSF WBC, OCB pos/ neg	MOG Status	symptoms	ophthal. findings, VA (Snellen, ft)	VEP	cMRI	initial therapy (listed chronologically)
Pat 1	c	43	124*; neg	clear 1:1280	vision loss, headache, fatigue	VA RE 20/100, LE 20/ 400, RAPD left, optic disc swelling $L > R$	no clear pathology	bilat WML subcortical*, (without Gad)	lumbar puncture, acetazolamide
Pat 2	c	NA	3; neg	low 1:10°	vision loss, headache	VA RE/LE between 20/40 and 20/100, bilat papilledema, decreased upward gaze eye movement and adduction left	RE/LE: prolonged P100 latencies* and lower amplitudes	normal (without Gad)	acetazolamide
Pat 3	a	38	201*; neg	low 1:10°	vision loss, headache, vomiting, diplopia	VA RE 20/100, LE 20/ 30, bilat papilledema with hemorrhage, abducent nerve palsy bilat, RAPD RE	NA	Signs for increased ICP: swollen perineural optic nerve sheath, signs of papilledema, flattened pituitary gland, (with Gad)	lumbar puncture, acetazolamide, IVIG
Pat 4	c	60	0; neg	clear 1:100°	vision loss, headache, vomiting, diplopia	bilat papilledema with hemorrhage, abducent nerve palsy, loss of vision (VA NA)	NA	signs of increased ICP: undulated optic nerve bilat, swollen optic nerve sheath*, flattened bulbus at optic disc entry, flattened pituitary gland, (with Gad)	acetazolamide
Pat 5	a	50	46*; neg	clear 1:2560	vision loss, bulbar movement pain*, decreased color saturation	severe bilat papilledema, VA LE/ RE 20/400	RE/LE: significantly prolonged P100 latencies*	Normal, (with Gad)	lumbar puncture, ventriculo- peritoneal shunt
Pat 6	a	34	4; neg	low 1:160°	vision loss, headache, nausea, bulbar movement pain*, decreased color saturation	VA RE/LE 20/125, bilat papilledema with hemorrhages, LE RAPD	RE: normal, LE: prolonged P100 latency* (134ms)	LEON left*, (with Gad)	lumbar puncture, IVMP
Pat 7	c	35	5; neg	low positive, titer not reported	vision loss, headache, vertigo, decreased color saturation	VA RE/LE 20/200, bilat papilledema	RE/LE: mild increase in P100 latency*, RE: mild decrease in amplitude	normal, (without Gad)	acetazolamide
Pat 8	a	54	53*; neg	clear 1:10,240	vision loss, headache, nausea, vertigo	VA RE 20/63, LE 20/ 32, bilat papilledema	LE: prolonged P100 latency*; RE: no detectable potential*	discrete enhancement optic nerve*, (with Gad)	acetazolamide

Table 1: Clinical and demographic data including CSF -, ophthalmological findings, visual evoked potentials (VEP), MRI findings and therapy at acute presentation are summarized. MOG test results are graded in clear positive titer (live CBA $\geq 1:640$; Euroimmune $\geq 1:100$) and low titer (live CBA 1:160–320; Euroimmune 1:10). °fixed CBA (Euroimmune); *findings are not typical for (fulminant) IIH and alert to further diagnostic work-up. Patients are grouped by age: child 0–12 years, adolescent: 13–17 years.

Abbreviations: CSF – cerebrospinal fluid, f - female, Gad – Gadolinium, ICP – intracranial pressure, IVIG – intravenous immunoglobulin, IVMP – intravenous methylprednisolone, LE – left eye, LEON – longitudinal extensive optic neuritis, m - male, NA – not available, OCB – oligoclonal bands, ON – optic neuritis, RAPD - relative afferent pupillary defect, RE – right eye, VA – visual acuity, WBC – white blood cell count, WML – white matter lesion.

Retrospectively, in all patients diagnosis of optic neuritis was based on clinical symptoms (bulbar movement pain, decreased color desaturation, RAPD), clinical findings (prolonged P100 latencies in VEP) and/or initial or subsequent MRI findings.

Serum MOG testing was performed using live and fixed CBA during initial workup or in the latter stage of the disease because of insufficient improvement. Results were positive in all eight children, in fixed cell-based assays (CBA; $n = 4$; range: 1:10 – 1:100) and live CBA ($n = 4$; range: 1:160 – 1:10,240).

White blood cell count (WBC) in CSF was normal in four children. In four children CSF cell count was elevated with a median of 88/ μ l (range: 0–201/ μ l). Oligoclonal bands (OCB) were negative in all patients.

MRI was performed in all patients during the first four days after admission. 3/8 patients had normal MRI findings without signs of increased intracranial pressure or inflammation. Signs for ON were

described in two patients: In patient 7 a longitudinal extensive ON (LEON) on the left side was noted, and in patient 8 discrete enhancement of the optic nerves. One patient had small unspecific, T2-hyperintense lesions bilaterally in the left frontal and parietal cortex at the initial presentation.

In two patients, signs for intracranial pressure were noted: Patient 3 and 4 had swollen perineural optic nerve sheaths, a tortuous course of both optic nerves and signs of papilledema in addition to an empty sella. Information about applied contrast medium can be seen in Table 1.

Subsequently, in patient 5 a longitudinal extensive transverse myelitis (LETM) was described in the cervical area. In patient 2 hyperintensities of the optic nerve were observed subsequently, not present in the initial MRI. Patient 4 developed bilateral white matter lesions and cervical LETM, compatible to new clinical symptoms with ataxia and back pain 16 days after the initial presentation.

4. Optical coherence tomography results

In 4/8 patients repetitive OCT data, starting from the acute stage of the disease, were available. To analyze OCT measurement in children with an acute MOG-positive bilateral ON, two further OCT data from patients with MOG-positive bilateral ON and optic disc edema were available. These two patients were not included in our clinical case series, as they were not diagnosed with IIH at initial presentation. We included these two patients in analysis of OCT measurement, to show that development of OCT findings was similar over time. OCT confirmed a substantial papilledema at baseline in all six patients. Pronounced initial global pRNFL swelling of median 256 μm (IQR: 136–485 μm) decreased rapidly within the first two weeks (–42 %) and between weeks 2 and 4 (further thinning of –25 %). Starting from week 4 a significant decrease of pRNFL thickness compared to baseline could be documented (Fig 1). Three months after onset all patients developed global pRNFL atrophy (<94 μm , median 68.5 μm , IQR 41–91 μm).

7/8 patients were treated with pressure-lowering therapies: 6 patients received acetazolamide. In two further patients a temporary CSF external drainage was performed. In one of the two latter children a VP shunt was implanted subsequently in view of persisting symptoms and a markedly elevated CSF opening pressure of 50 cm H₂O. IVIG was given in one patient in addition to acetazolamide, because of insufficient improvement of symptoms. One patient was transferred to another hospital for treatment of IIH, however, diagnosis was adapted to MOGAD and IVMP was started.

In all patients FU data were available with a median of 9.5 months (range: 3–30 months). In six patients, symptoms improved completely after treatment with i.v. steroids. One patient still has visual residuals with blurred vision and pathologic VEP results after 30 months. Additionally, she developed migraine-like headaches and a mild right-sided weakness which improved but was still present after 30 months. In patient 8 with a short follow-up of 3 months, symptoms decreased significantly but a final visual assessment is pending.

5. Case report

5.1. Patient 5

The 14-year-old boy presented to the emergency department because of rapid visual loss over few days. Examination showed an obese boy (BMI 29.4 kg/m²) with bilateral loss of vision and mild headache

but no focal neurological deficits. Ophthalmological examination showed a nearly complete loss of vision, with significantly prolonged P100 latencies in VEP examination, and severe optic disc swelling on both eyes. Cerebral MRI (with contrast medium) showed only signs of bilateral papilledema but no further pathologies. Lumbar puncture showed an elevated opening pressure of 54 cm H₂O together with a cell count of 46 leukocytes/ μL , OCBs and serological testing for *Borrelia burgdorferi* and Herpes Simplex virus (HSV) were negative. Fulminant IIH was suspected and treatment was initiated with the insertion of a VP shunt and oral acetazolamide. Visual acuity in both eyes did not improve over the following days. A spinal MRI was performed several days later because of evolving bladder dysfunction, showing a LETM in the cervical region. The combination of an elevated CSF cell count and LETM suggestive of inflammatory CNS disease led to the administration of IVMP for 5 days with rapid improvement of symptoms and discharge. Serum was tested for MOG- and AQP4 abs revealing a high MOG-ab titer of 1:2560. Low-dose oral prednisolone therapy of 10 mg every second day was continued for 4 months. 24 months after the acute episode, visual acuity and cMRI were normal. MOG abs were tested negative and his VP shunt was surgically removed after a 12 months period of slowly increasing the pressure setting.

Fig. 1: Peripapillary retinal nerve fiber layer (pRNFL) thickness change over time after optic neuritis.

6. Discussion

In this retrospective case series, we report 8 children who were initially diagnosed with fulminant IIH and treated accordingly including oral acetazolamide, regular CSF lumbar punctures and even a VP shunt insertion in one patient. In all children MOG-abs were found leading to the final diagnosis of MOGAD with bilateral ON and the appropriate immunomodulatory treatment.

Progressive visual loss can be found in a variety of rare diseases (Wendel et al., 2020; Bouffard, 2020; Bennett, 2019; Lenaers et al., 2023; Dammacco et al., 2022). In particular fulminant IIH is associated with rapid visual loss in conjunction with headache and raised CSF opening pressure. (Bouffard, 2020) Another differential diagnosis is MOGAD with bilateral ON which can present also with rapid central acuity loss and retro-orbital pain often described as severe headache. (Asseyer et al., 2020) Interestingly, optic disc swelling seen on fundoscopy is common with MOGAD optic neuritis, which can be mistaken for papilledema from raised intracranial pressure. A typical MRI feature of

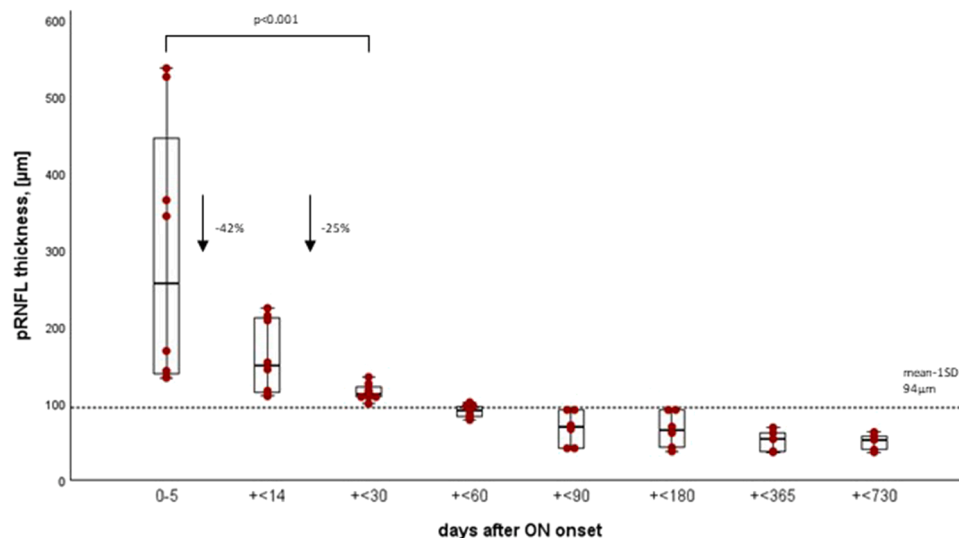


Fig. 1. Substantial papilledema and pRNFL thickening have been observed during the first five days since onset in all six patients (4/6 initially diagnosed as IIH, 2/6 as MOGAD). Rapid decrease of pRNFL followed already within the first 14 days after ON onset, leading to global pRNFL atrophy (all eyes below 94 μm) in the first three months.

MOGAD-ON is the inflammation of the anterior, pre-chiasmatic part of the optic nerve with longitudinal involvement termed LEON. (Wendel et al., 2020) Therefore MOGAD-bil ON has clearly overlapping features in the clinical presentation with fulminant IIH in particular in the presence of severe bilateral papilledema.

All reported children in this case series did present initially with symptoms highly suggestive of fulminant IIH. All children had headache and significant and rapid visual impairment in both eyes, partly with preceding signs such as loss of color vision or pain on eye movement. The diagnosis of fulminant IIH was further supported by clinical findings such as nausea, emesis and 6th nerve palsy in a subgroup of children. Work-up showed a markedly elevated CSF opening pressure in 7/8 patients, again pointing to the diagnosis of fulminant IIH.

In general, fulminant IIH is rare in children. In a population-based study from Germany not a single pediatric fulminant IIH case was reported during a one-year period. (Tibussek et al., 2013). With an estimated annual incidence of 150 pediatric PTCS cases in Germany (Tibussek et al., 2013), around five pediatric fulminant IIH cases per year can be assumed. (Thambisetty et al., 2007) Therefore, if the clinical course suggests a fulminant IIH, thorough evaluation of differential diagnosis is always warranted, in order to not miss other treatable conditions and to prevent unnecessary invasive procedures. It has been shown in the past that pediatric PTCS is overdiagnosed for several reasons (Mishra et al., 2007). First, the recommended diagnostic criteria are not properly applied, (Friedman et al., 2013) e.g. CSF pleocytosis is a finding that should exclude IIH (Table 2). Secondly, differentiation of papilledema due to intracranial hypertension from optic disc swelling due to other etiologies is challenging. Thirdly, an opening pressure above normal does not necessarily result from a sustained increase in intracranial pressure. In children in whom CSF opening pressure is recorded greater than 28 cm H₂O, interpretation of the result should also include clinical findings suggestive of PTCS and information about the lumbar puncture, because several factors do influence the CSF opening pressure (e.g. sedation, BMI, agitated patient during procedure). (Avery, 2014)

More recently, several studies reported an increase in CSF opening pressure in children and adults with ADS. Up to one-third-of children with demyelinating CNS diseases like ADEM can have an increased CSF opening pressure. (Narula et al., 2015; Morgan-Followell and Aylward, 2017; Nagaratnam et al., 2024; Tisavipat et al., 2024) The pathomechanisms leading to the increased CSF opening pressure are not fully understood. Impaired CSF absorption or dysregulation of CSF flow dynamics due to inflammatory processes are possible explanations. (Narula et al., 2015; Morgan-Followell and Aylward, 2017).

In a recently published study 43/86 MOG-positive children had intracranial hypertension (>28 cm H₂O), with a higher percentage of ADEM patients compared to patients with ON, myelitis or other forms of MOGAD. Patients with ADEM had a significantly higher opening pressure (median: 30 cm H₂O) than patients with ON (median: 20.5 cm H₂O). (Nguyen et al., 2023) In our cohort of MOGAD-ON patients,

median opening pressure was significantly higher with 48.5 cm H₂O. The significantly elevated opening pressure in this subgroup indicates, that CNS inflammation can lead to a secondary intracranial hypertension with potential prognostic impact on vision loss and outcome. In addition to treatment of the CNS inflammation, CSF pressure management should be discussed as suggested recently by Ngyuen et al., who observed a longer stay in hospital and greater likelihood of neurologic disability in MOGAD-ADEM patients with elevated opening pressure. (Nguyen et al., 2023) Three patients in our cohort did receive regular spinal taps or even a shunt insertion leading to improvement of symptoms. At present we can only speculate if prompt treatment with IVMP would have been suffice.

To date, there are no data with regard to the impact of intracranial hypertension on visual outcome in MOGAD-ON patients. In principle, the majority of patients with MOGAD-ON shows good response to i.v. steroids with restitutio ad integrum. (Chen et al., 2018)

Female gender and obesity are regarded as risk factors for PTCS foremost in post-puberty IIH. (Friedman et al., 2013) Interestingly, 7/8 patients in our cohort were female, only 1/8 patient was obese. This observation raises the question, if female gender also plays a role in developing intracranial hypertension in neuroinflammatory processes. In a retrospective cohort study of 53 children with demyelinating CNS disease no association between female gender and BMI with increased intracranial pressure was found. (Narula et al., 2015) The latter can be supported by our findings, as the median BMI of our cohort was 18.15kg/m². As shown above in patient 8, especially in obese post-pubertal patients the diagnosis of MOGAD-bilON can be difficult, and the presence of clinical red flags (Table 2) can be helpful in the differential diagnosis.

The following clinical features could have alerted the treating physician to an inflammatory cause (see also Table 2). First, an elevated CSF cell count was reported in four children. Elevated WBC in PTCS is very untypical (Morgan-Followell and Aylward, 2017) mandating the exclusion of a range of neuroinflammatory diseases. Secondly, in 2/3 patients of our cohort with normal CSF findings, the patients described typical symptoms of ON like bulbar movement pain and decreased color saturation. Clinical hints like optic disc swelling in MOG-positive ON should be kept in mind to include MOG-IgG testing in the differential work up. Differentiation of papilledema and optic disc swelling is challenging, even with available OCT data. Pathological VEP latencies, as observed in 5/6 patients in this cohort, or RAPD speaks against IIH and are suggestive of ON as underlying cause of visual impairment. Thirdly, rapid decline of pRNFL thickness observed in our cohort already during two weeks after onset seems to be rare in IIH and is more characteristic for MOGAD. (Chen et al., 2022)

The following limitations have to be addressed. Firstly, clinical findings and several investigations such as ophthalmological examination, CSF opening pressure, OCT were not obtained in standardized fashion. Secondly, reliable evaluation of an association between CSF pressure management (spinal tap, acetazolamide) and improvement of IIH symptoms (e.g. headache) was not possible. Thirdly, original OCT data were only available from four patients of this cohort. We included two further original OCT data from children with MOGAD-bilON, presenting with vision loss and papilledema, who were promptly diagnosed with MOGAD and hence could not be included in our case series. However, this does not influence the comparability of OCT data. Lastly should be mentioned that different test methods were used to detect and measure serum MOG-ab titers including fixed and live cell-based assays.

7. Conclusion

Children with bilateral MOGAD-ON presenting with optic disc swelling and loss of vision are at risk of being misdiagnosed as fulminant IIH. Recognizing the critical signs in order to make the correct diagnosis is fundamental in terms of treatment implications. Apart from MR-imaging of the orbita, we strongly recommend to include

Table 2
Red flags in the differential diagnosis of increased intracranial hypertension.

Symptoms	rapid decrease of visual acuity bulbar movement pain decreased color saturation
CSF studies	increased WBC
Ophthalmological examination	RAPD
VEP	prolonged P100 latency
OCT	rapid pRNFL decrease
MRI	pronounced optic nerve sheath contrast enhancement optic nerve white matter lesions

Abbreviations: CSF – cerebrospinal fluide, MRI – magnetic resonance imaging, OCT – optical coherence tomography, pRNFL – peripapillary retinal nerve fiber layer, RAPD - relative afferent pupillary defect, VEP – visual evoked potentials, WBC – white blood cell count.

measurement of CSF opening pressure in the standard work up of neuroinflammatory diseases and in particular in patients with suspected MOGAD, in order to gain more insights in CSF dynamics in this disease and to consider adapted pressure management if needed.

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CRedit authorship contribution statement

Eva-Maria Wendel: Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Conceptualization. **Daniel Tibussek:** Writing – review & editing, Methodology, Data curation. **Nina Barisic:** Writing – review & editing, Data curation. **Annikki Bertolini:** Writing – review & editing, Data curation. **Andreas Panzer:** Writing – review & editing, Resources. **Petrus Chang:** Writing – review & editing, Conceptualization. **Tobias Geis:** Writing – review & editing, Data curation. **Ellen Knierim:** Writing – review & editing, Data curation. **Marc Nikolaus:** Writing – review & editing, Data curation. **Margherita Nosadini:** Writing – review & editing, Data curation. **Stefano Sartori:** Writing – review & editing, Data curation. **Jan-Christoph Schoene-Bake:** Writing – review & editing, Data curation. **Deniz Yilmaz:** Writing – review & editing, Data curation. **Markus Reindl:** Writing – review & editing, Resources, Investigation, Data curation. **Thivya Pakeerathan:** Writing – review & editing, Investigation, Data curation. **Ilya Ayzenberg:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Kevin Rostásy:** Writing – original draft, Supervision, Data curation, Conceptualization.

Declaration of competing interest

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